

HOMEWORK 2: GRAPH ALGORITHMS

1/ Definitions of the following concepts:

(I) K -edge-coloring: A K -edge-coloring of Graph $G = (V, E)$ is a mapping $C: E \rightarrow \{1, \dots, K\}$ such that for any 2 edges $e, f \in E$ sharing a common endpoint, $C(e) \neq C(f)$.

(II) G is K -edge-colorable: A graph G is K -edge-colorable if there exists a valid K -edges-coloring of G .

(III) The Chromatic Index of G denoted $\chi'(G)$ is the smallest integer K such that G is K -edge-colorable.

b) According to Vizing's theorem $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$ where $\Delta(G)$ is the maximum degree of G . G cannot be edge-colored with fewer than $\Delta(G)$ colors because if a vertex v has degree $\Delta(G)$ there are $\Delta(G)$ edges incident to it.

In any valid edge coloring all edges incident to a single vertex must receive distinct colors.

therefore, at least $\Delta(G)$ colors are required.

c) let $K = \Delta(G) \geq 2$. Algorithm A produces a $(K+1)$ -edge-coloring

- the number of colors used by the algorithm is $SOL = K+1$
- the optimal number of colors is $OPT = \chi'(G)$

We know $OPT \geq \Delta(G) = K$.

the approximation ratio ρ is: $\rho = \frac{SOL}{OPT} \leq \frac{K+1}{K} = 1 + \frac{1}{K}$.

Since $K \geq 2$ the maximum value of $1 + \frac{1}{K}$ is $1 + \frac{1}{2} = \underline{\underline{1.5}}$.

therefore A is a $(3/2)$ -approximation algorithm.

2/ Improved Greedy ^{coloring} Algorithm.

a) degeneracy 5: the improved greedy algorithm uses most $d+1$ colors where d is the degeneracy of the graph.

• Here degeneracy $d = 5$ so the algorithm uses at most $5+1 = 6$ colors.

- if a graph has degeneracy 5 it contains a subgraph with minimum degree 5 which implies the graph has edges. Since the graph has edges, its chromatic number $\chi(G) \geq 2$.

the approximation ratio is $\frac{6}{\chi(G)} \leq \frac{6}{2} = 3$, thus it's a 3 approximation.

b) Degeneracy K :

- The algorithm uses at most $K+1$ colors
 - Assuming the graph has at least one edge (which is true if $K \geq 1$), $\chi(G) \geq 2$.
- the approximation ratio is $\frac{K+1}{2}$.

c) Show that if G is a Tree, then ~~the~~ the improved greedy coloring optimally colors G .

- Trees have a degeneracy of $K = 1$ (every subgraph of a tree containing a vertex of degree 1).
- the algorithm uses at most $1+1 = 2$ colors.
- Trees with at least one edge are bipartite, so $\chi(\text{Tree}) = 2$ (if edgeless $\chi = 1$, algorithm uses 1).

Since the algorithm uses $\chi(G)$ colors, it colors G optimally

3) Vertex cover approximation

a) Let $A(G)$ be the output of the approximation algorithm and C^* be a minimum vertex cover, the inequalities are: $|C^*| \leq |A(G)| \leq 2|C^*|$

b) True. $A(G)$ is a vertex cover, meaning it covers all edges. The set $B(G) = V \setminus A(G)$ contains no two vertices connected by an edge (otherwise that edge would not be covered by $A(G)$). Thus $B(G)$ is an independent set.

c) False. Considering K_n, n . the maximum independent set size is n . The algorithm A (based on maximal matching) might select all $2n$ vertices as the cover. Then $B(G)$ would return an empty set (size 0). The ratio $n/0$ is not constant.

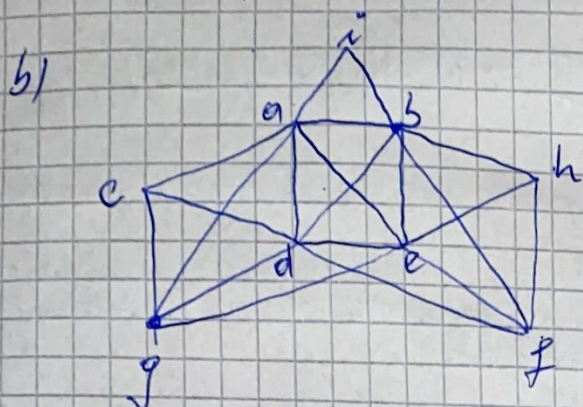
d) False. Let G be P_3 (1-2-3). Choose $I = \{1\}$ and $C = \{2\}$ both have size 1. C is a minimum vertex cover, but I is not a maximum independent set (optimal is $\{1,3\}$ size 2). Equality of size does not imply optimality for the independent set.

e) True. if an algorithm is a c -approximation, $Sol \leq c \cdot OPT$ if $d > c$, then $Sol \leq c \cdot OPT < d \cdot OPT$.

4) a) 1. A chordal graph is a graph where every cycle of length greater than 3 has a chord.

2) A simplicial vertex is a vertex whose neighbors form a clique (i.e. every pair of neighbors of v is connected by an edge).

3) A perfect elimination ordering is an ordering of vertices such that each vertex is simplicial in the graph induced by itself and the subsequent vertices.



The simplicial vertices are:

- c : Neighbors $\{a, d, e\}$
- h : Neighbors $\{b, c, d, e\}$
- i : Neighbors $\{a, b\}$

c) Perfect elimination ordering using the rule "choose alphabetically smallest"

1) the simplicial candidates: $\{c, h, i\}$. I choose c
 - Graph remaining $G - \{c\}$, vertex d (neighbors a, e, g) becomes simplicial

2) Candidates $\{d, h, i\}$, I choose d , Graph remaining $G - \{c, d\}$

3) Candidates $\{h, i\}$, (vertices a, g are not simplicial yet) I choose h
 - Graph remaining $G - \{c, d, h\}$, vertex f (neighbors b, e, g) becomes simplicial

4) Candidate $\{i\}$:
 - Graph remaining $G - \{c, d, h, i\}$ vertices $\{a, b, e, f, g\}$
 a becomes simplicial (neighbors b, e, g) ($b-g$ is the chord)

5) Candidates $\{a, f\}$, I choose a
 - Graph remaining $G - \{c, d, h, i, a\}$ vertices $\{b, e, f, g\}$

6) Candidate ~~a~~ $\{b, f\}$, I choose b .
 - Graph remaining $G - \{c, d, h, i, a, b\}$ vertices $\{e, f, g\}$

7) Candidates $\{e, f, g\}$ I choose e .
 - Graph remaining $G - \{c, d, h, i, a, b, e\}$ vertices $\{f, g\}$

8) Candidates $\{f, g\}$, I choose f , then g .

9) the ordering: $c, d, h, i, a, b, e, f, g$

d) using the ordering above, I would apply greedy coloring in the reverse order of the PEO $\Rightarrow \{g, f, e, b, a, i, h, d, c\}$.

$\Rightarrow$ g : color 1 - f : (neighbors g) color 2 - e : (Neighbors g, f) color 3
 - b : (neighbors g, f, e) color 4, - a : neighbors g, e, b , color 2
 - i : neighbors a, b : color 1
 - h : neighbors b, f : color 1
 - d : neighbors a, e, g : color 4
 - c : Neighbors a, d, g : color 3

$$\chi(G) = 4.$$

4) Maximum Independent set of a chordal graph using PEO.

Method: Iterating through the PEO, if a vertex v is available, add it to I and discarding its neighbors
PEO: $c, d, h, i, a, b, e, f, g$.

1) c : add to I , $\{a, d, g\}$ discarded.

d : discarded

h : Add to I , $\{b, f\}$ discarded

i : Add to I , $\{a, b\}$ discarded.

a, b, e, f, g : All discarded, then $I = \{c, h, i\}$ size 3.

5) Dirius theorem: A graph is an interval graph $\Leftrightarrow$ it has an interval order.

$\Rightarrow$ We know a graph is chordal $\Leftrightarrow$ it has a perfect elimination ordering.
Therefore since every interval graph has a PEO (via its interval order) every interval graph is chordal.

5) a) Minimum number of colors (coloring): Sorting groups by start time (left endpoint)

$G_3(08:00), G_7(09:00), G_2(10:30), G_6(11:00), G_{10}(12:00), G_9(12:30)$
 $G_8(13:30), G_1(14:30), G_5(15:30), G_4(16:30)$

$\rightarrow$ Greedy coloring:

1) $G_3(08:00 - 10:00)$: Color 1

2) $G_7(09:00 - 11:30)$: Color 2

3) $G_2(10:30 - 13:00)$: Color 1 (G_3 free)

4) $G_6(11:00 - 14:00)$: Color 3 (overlaps G_7, G_2)

5) $G_{10}(12:00 - 17:00)$: Color 2 (G_7 free)

6) $G_9(12:30 - 15:00)$: Color 4 (overlaps G_2, G_6, G_{10})

7) $G_8(13:30 - 18:00)$: Color 1 (G_2 free)

8) $G_1(14:30 - 16:00)$: Color 3 (G_6 free)

9) $G_5(15:30 - 20:00)$: Color 4 (G_9 free)

10) $G_4(16:30 - 19:00)$: Color 3 (G_1 free)

the minimum number of colors = 4

5) the maximum independent set using DP on interval order (sorted by finish time), ~~the max~~ and calculating $(p[i])$ last non overlapping

1) $G_3[7]$ ($p=0$) 3) $G_2[10]$ ($p=G_3$)

2) $G_7(10)$ ($p=0$) 4) $G_6(8)$ ($p=G_3$)

5) $G_9(14)$ ($p=G_7$) 8) $G_8(15)$ ($p=G_2$)

6) $G_{10}(16)$ ($p=G_6$) 9) $G_4(12)$ ($p=G_1$)

7) $G_{10}(8)$ ($p=G_7$) 10) $G_5(9)$ ($p=G_9$)

using dynamic programming $DPT(i) = \max(w_i + OPT(p(i)), OPT(i-1))$

We have $G_3: 7$

$G_7: \max(10+7) = 17$

$G_2: \max(10+7, 10) = 17$

$G_6: \max(8+7, 17) = 17$

$G_9: \max(14+10, 17) = 24$

$G_{10}: \max(16+10, 24) = 24$

$G_8: \max(15+17, 24) = 32$

$G_1: \max(12+24, 32) = 36$

$G_5: \max(9+24, 36) = 36$

$G_4: \max(12+24, 36) = 36$

the max tourist: 36

Groups selected: 7, 9, 4.